

A380 takeoff calculator and company T.O data

Takeoff weight limit, FLEX temperature, V-speeds and takeoff run on the flyPad, sent to the MFD PERF T.O page through the company takeoff data function of the FCOM

Aircraft	A380X (the flyPad screen is shared with the A32NX, see the PR 12 manual)
Where to find it	flyPad > Performance > Takeoff ; MFD > ACTIVE > PERF > T.O tab > CPNY T.O REQUEST / RECEIVED CPNY T.O
PR branch	pr/11-a380-takeoff-calculator (stacked on PR 10); completed by PR 12
Status	Used in the sim during development (September 2026); the final version with the PR 12 changes is still to re-test in the sim.
Pictures	flyPad and MFD pages in the local browser harness, running the build installed in the sim. The harness has no navigation database: the runway is entered by hand and the company uplink is simulated on the MFD.

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1. What it is for

Before each takeoff the crew needs the takeoff weight limit of the runway, the FLEX temperature (reduced thrust), the speeds V1, VR, V2 and the flaps. On the real A380 they come from the OIS takeoff application or from the airline (company takeoff data, sent to the FMS by datalink). The flyPad calculator does the computation; SEND TO FMS plays the airline: the MFD receives the data as company T.O data, and the crew checks and inserts it, as in FCOM DSC-22-FMS-20-30.

2. The flyPad screen

Mon Sep 281024Z

100%

Performance

TakeoffTop of DescentLandingTemperature Correction

Airport

ICAO

Fill data from

FMS

RUNWAY

RunwayManual

Heading

TORAm

Elevationft

Slope%

Entry angle90°

CONDITIONS

ConditionDry

Wind°/kt

OAT°C

QNHhPa

Anti-IceOff

PacksOn

AIRCRAFT

TOWkg

T.O CG% MAC

FlapsCONF 1+F

ThrustFLEX

DEPARTURE

THR REDft

ACCELft

EO ACCELft

Noise procedure

NOISE ENDft

SPDkt

THR N1%

RESULTS

TOWt

V1kt

VRkt

V2kt

T.O CG%

THR REDft

MTOW(perf)t

MARGINm

FLAPS

T.O SHIFTm

ACCELft

EO ACCELft

TAKEOFF RUN

ThrustTOGAFLEX

Calculate to see the takeoff run

Calculate

Clear

Send to FMS

Required: Airbus take-off weight charts (all limitations and margins). V1, VR, 35 ft: estimates, all engines.

Figure 1. The takeoff screen. Amber labels: inputs still missing (Calculate stays dim).

Part	Contents
Airport, Fill data from	ICAO code; imports from the FMS (departure, runway, TOW, CG, THR RED / ACCEL), the OPF (SimBrief) or the METAR (wind, OAT, QNH).
RUNWAY	Runway from the list (or Manual), heading, TORA, elevation, slope, and the entry angle (90° or 180° turn onto the runway, or 0° when rolling straight on: it sets the line-up distance lost).
CONDITIONS	Runway condition, wind, OAT, QNH, anti-ice, packs.
AIRCRAFT	TOW in t (or klb), one decimal, T.O CG in % MAC, flaps (CONF 1+F, 2, 3 or OPT = the best one), thrust (TOGA or FLEX).
DEPARTURE	THR RED, ACCEL, EO ACCEL altitudes (they default to the elevation + 1500 ft), and the noise abatement procedure (NOISE END altitude, speed, N1).
RESULTS	The results in the layout of the OIS takeoff application (FCOM PER-TOF-TOR-SRS).
TAKEOFF RUN	The runway with the V1, VR and 35 ft points, the required distance and the margin, for TOGA or any FLEX temperature of the allowed range.

3. A FLEX takeoff

1. Fill the airport and runway (or **Fill data from FMS / OPF**), the conditions (METAR), the TOW and the T.O CG.
2. Keep **Thrust FLEX** and press **Calculate**.
3. Read the FLEX temperature and the speeds in RESULTS; check the takeoff run and its margin.
4. **Send to FMS**, then insert the data on the MFD (section 6).

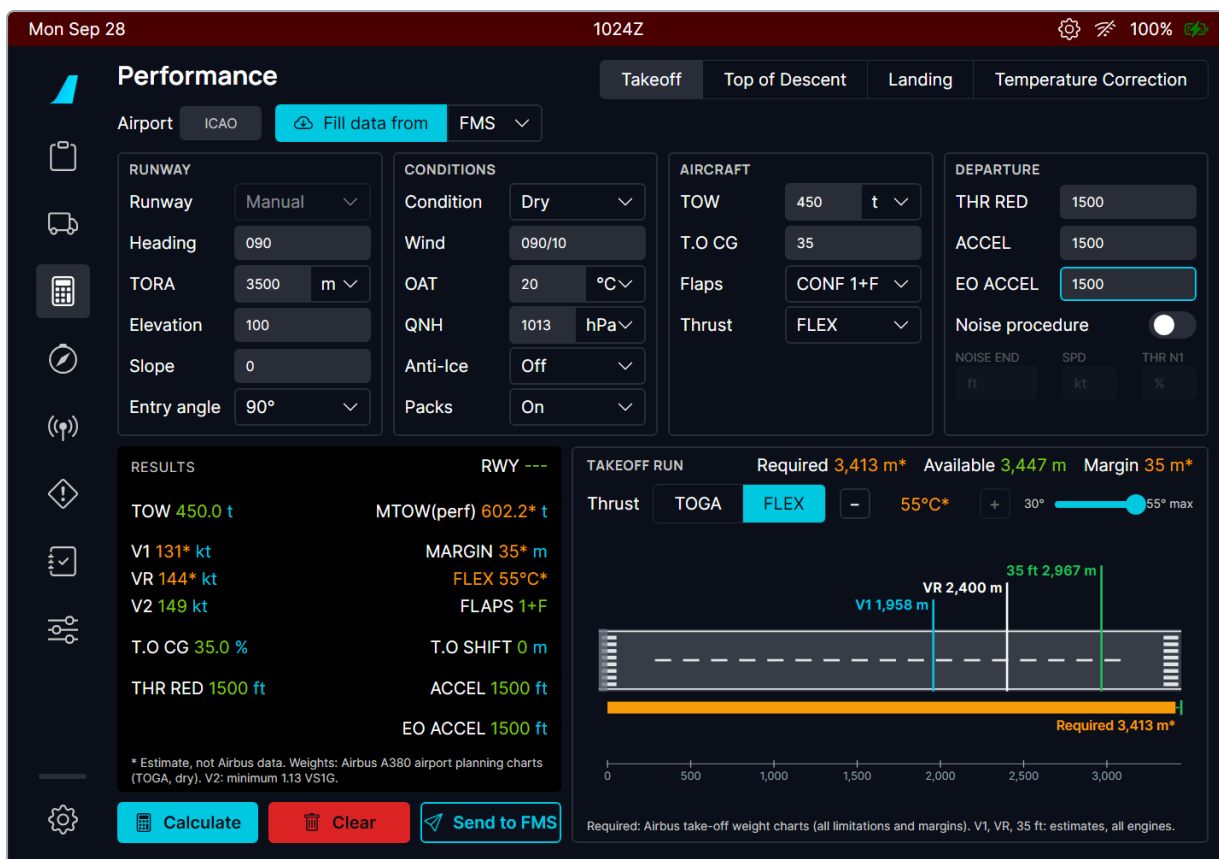


Figure 2. 450 t, 3500 m, 20 °C, CONF 1+F: FLEX 55 °C (the maximum, amber: estimate), V1 131 / VR 144 / V2 149 kt, MTOW(perf) 602 t. The takeoff run needs 3413 m of the 3447 m available after line-up: 35 m margin.

Reading the results

Line	Meaning
TOW / MTOW(perf)	Your takeoff weight, and the maximum the runway and conditions allow at TOGA.
V1, VR, V2	The takeoff speeds. V2 is at least 1.13 VS1G and 1.10 VMCA, VR at least 1.05 VMCA, V1 at least VMCG (FCOM PER-TOF-TOR-SRS).
MARGIN	Runway left beyond the required distance.
FLEX / TOGA	The FLEX temperature, or TOGA.
FLAPS, T.O CG, T.O SHIFT	The configuration, the CG entered, the takeoff shift (distance lost at line-up or an intersection).
THR RED, ACCEL, EO ACCEL	The altitudes of the DEPARTURE section.

NOTE

Green values come from Airbus data (the takeoff weight charts of the A380 Aircraft Characteristics); **amber with a star** are estimates of the FBW model (see Limitations).

4. TOGA or another FLEX temperature

The TAKEOFF RUN panel lets you look at the run for TOGA, or for any FLEX temperature between the OAT and the maximum (FCOM PER-TOF-THR-FLX). Use - / + or the slider; the speeds and the run follow.

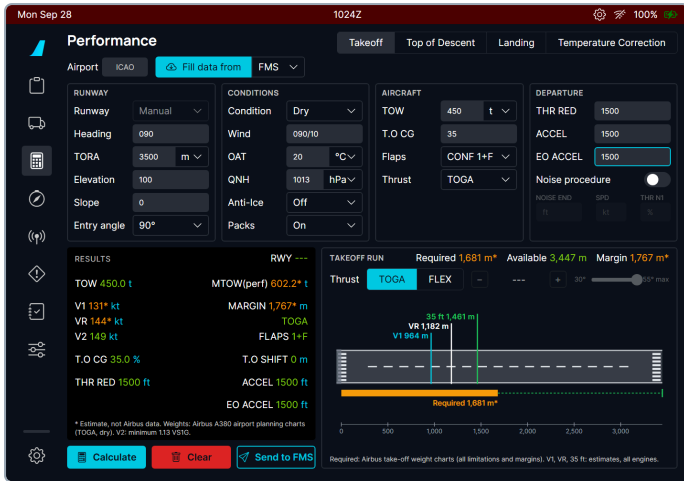


Figure 3. TOGA: the run needs 1681 m, margin 1767 m.

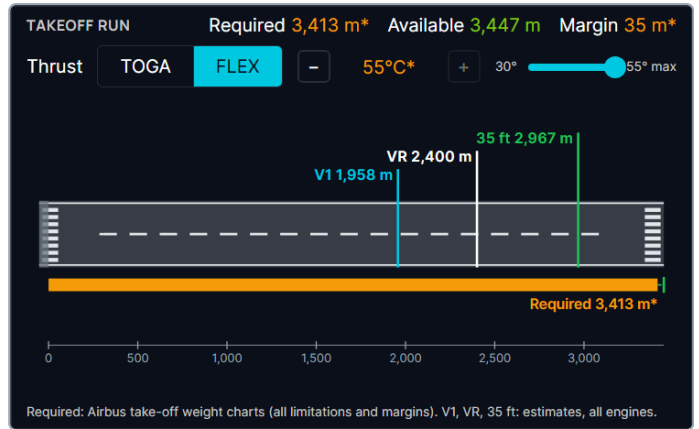


Figure 4. FLEX range 30 °C to 55 °C max: the selected temperature is the one sent to the FMS.

Send to FMS sends the thrust selected for the takeoff run only (TOGA, or FLEX at its temperature), with its own speeds.

5. Options

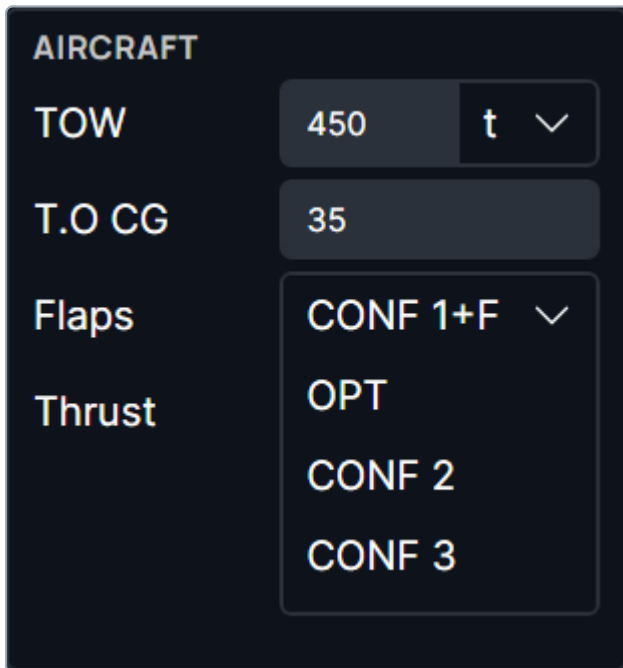


Figure 5. Flaps: CONF 1+F, 2, 3, or OPT (the calculator picks the best).

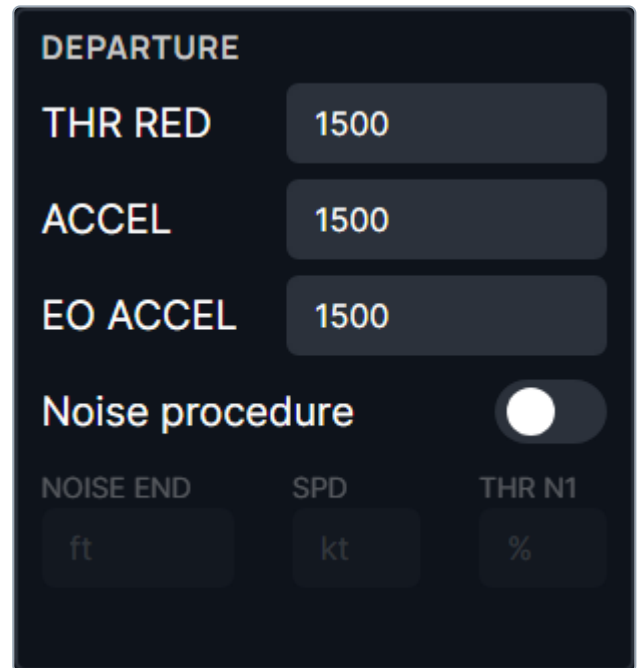


Figure 6. Noise procedure: NOISE END altitude, speed and N1 become available; they are sent with the data.

6. Errors

When the takeoff is not possible, a message says why and there is no result:

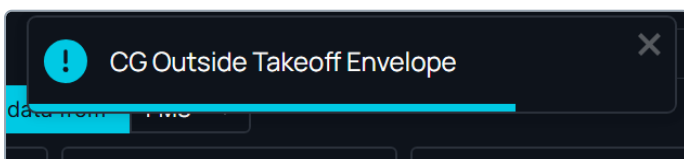


Figure 7. CG outside the takeoff envelope (here 30 % at 450 t: the forward limit moves aft with the weight).

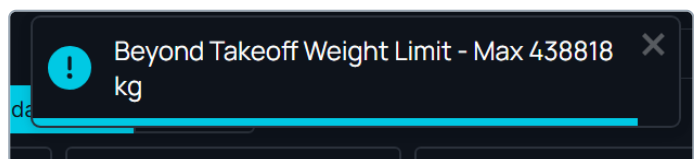


Figure 8. 510 t on 2100 m at 40 °C: the maximum takeoff weight is 438.8 t.

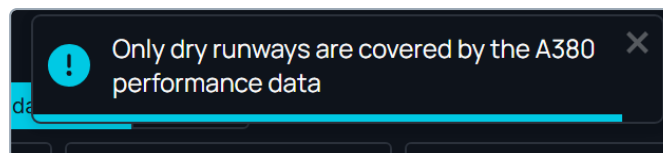


Figure 9. A wet or contaminated runway: the A380 data covers dry runways only.

Message	Meaning
Beyond Takeoff Weight Limit - Max ...	TOW above the runway limit: reduce the weight or use another runway.
CG Outside Takeoff Envelope	The CG is outside the A380 takeoff CG envelope for this weight.
Above Structural MTOW	TOW above 510 t.
Only dry runways are covered...	The A380 data has no wet or contaminated takeoff.
Runway shorter than the A380 performance data (1700 m)	The charts start at 1700 m.
Above Maximum Tailwind / OAT / Pressure Altitude	Outside the A380 limits (FCOM LIM-12).
Flex Not Possible	No FLEX temperature is possible at this weight: take off at TOGA.

7. On the MFD: company T.O data

The MFD PERF T.O page gets two buttons: **CPNY T.O REQUEST** (ask the airline) and, once data has arrived, **RECEIVED CPNY T.O.**

 The MFD PERF T.O page. At the top, it says "FMS 1" and "ACTIVE/PERF". Below that are tabs for "T.O", "CLB", "CRZ", "DES", "APPR", and "GA". The "T.O" tab is selected. It shows fields for "RWY", "T.O SHIFT", "V1", "VR", "V2", "F", "S", "O", "TOGA", "FLEX", "DERATED", "FLAPS", "THS FOR", "PACKS", "ANTI-ICE", "THR RED", "ACCEL", "NOISE", "TRANS", "EO ACCEL", and "CPNY T.O REQUEST". There are also "RETURN" and "POS MONITOR" buttons at the bottom.

Figure 10. PERF T.O before any data: the CPNY T.O REQUEST button (bottom right).

 The MFD COMPANY T.O DATA REQUEST page. At the top, it says "FMS 1" and "COMPANY T.O DATA REQUEST". Below that are fields for "REQUEST FOR RWY", "RWY 1/2", "T.O SHIFT", "T.O LIMIT", "TOW", "MAG WIND", "T.O CG", "RWY COND", "TEMP", "QNH", "TOGA", "FLEX", "DERATED", "FLAPS", "FREE TEXT", "SEND T.O REQUEST *", "RETURN", and "RECEIVED T.O DATA".

Figure 11. COMPANY T.O DATA REQUEST: up to 2 runways, TOGA / FLEX / DERATED, free text; SEND T.O REQUEST.

- After SEND T.O REQUEST the page shows *REQUEST PENDING...*; without an answer after 4 minutes, *NO COMPANY REPLY*. The flyPad **Send to FMS** is that answer.
- When data arrives: the message *COMPANY T.O DATA RECEIVED WAITING FOR INSERTION* and the button **RECEIVED CPNY T.O** on PERF.

FMS 1

ACTIVE POSITION SEC INDEX DATA

ACTIVE/PERF

CRZ FL 000 OPT FL --- REC MAX FL ---

T.O. CLB CRZ DES APPR GA

RWY --- T.O. SHIFT --- M

V1 000 KT F --- KT TOGA

VR 000 KT S --- KT FLEX

V2 000 KT 0 --- KT DERATED

FLAPS THS FOR PACKS ANTI-ICE

THR RED --- FT

ACCEL --- FT

NOISE

TRANS --- FT EO ACCEL --- FT

RECEIVED CPNY T.O

RETURN POS MONITOR

CLEAR INFO COMPANY T.O DATA RECEIVED WAITING FOR INSERTION

Figure 12. Data received: the message and the RECEIVED CPNY T.O button.

FMS 1

ACTIVE POSITION SEC INDEX DATA

RECEIVED COMPANY T.O DATA

DATA FOR RWY 05 FLEX 59 RWY 1/1

T.O. WEIGHT 371 T MAG WIND 058 KT 6 KT

T.O. CG 32.0% RWY COND DRY

TEMP --- °C QNH 1017

V1 119 KT TOGA

VR 131 KT FLEX

V2 136 KT DERATED

FLAPS 1 T.O. SHIFT --- M

T.O. LIMIT 8166 FT

THR RED 1530 FT

ACCEL 1530 FT EO ACCEL 1530 FT

NOISE END --- FT SPD --- KT

THR --- %

INSERT *

CLEAR *

T.O. PERF CPNY T.O REQUEST

CLEAR INFO COMPANY T.O DATA RECEIVED WAITING FOR INSERTION

Figure 13. RECEIVED COMPANY T.O DATA: runway 05 in amber (not the flight plan runway), TOW in amber with UPLINK/ACTIVE TOW DISAGREE: INSERT is not available.

The RECEIVED COMPANY T.O DATA page

- Up to 4 runways (arrows, RWY 1/4) × 2 thrust ratings (the TOGA / FLEX list at the top).
- **Runway in amber:** not the departure runway of the active flight plan.
- **TOW in amber** with *UPLINK/ACTIVE TOW DISAGREE*: the uplink TOW is more than 2 t below or 7 t above the FMS TOW (FUEL&LOAD).
- TEMP shows dashes for a FLEX takeoff; T.O. LIMIT is the TORA; the noise fields are filled when the procedure was sent.
- **INSERT** (only when runway and TOW agree) fills PERF T.O: speeds, flaps, thrust, shift, THR RED, ACCEL, EO ACCEL and noise. It does not fill THS FOR: the crew enters the CG there. **CLEAR** deletes the data.

FMS 1

ACTIVE

POSITION

SEC INDEX

DATA

RECEIVED COMPANY T.O DATA

DATA FOR RWY 05

FLEX 59

◀

▶

RWY 1/1

T.O WEIGHT 371 T

MAG WIND 058 °

6 KT

T.O CG 32.0%

RWY COND DRY

TEMP --- °C

QNH 1017

V1 119 KT

VR 131 KT

V2 136 KT

FLAPS 1

TOGA

FLEX

DERATED

T.O SHIFT --- M

T.O LIMIT 8166 FT

THR RED 1530 FT

ACCEL 1530 FT

EO ACCEL 1530 FT

NOISE END 3000 FT

SPD 180 KT

THR 85 %

INSERT *

CLEAR *

T.O PERF

CPNY T.O REQUEST

CLEAR INFO

COMPANY T.O DATA RECEIVED
WAITING FOR INSERTION

Figure 14. An uplink with the noise procedure: NOISE END 3000 ft, SPD 180 kt, THR 85 %.

TIP

The flyPad warns you right after Send to FMS if INSERT will not work: another departure runway in the FMS, no TOW in the FMS, or a TOW outside -2 t / +7 t of the FMS one.

8. What to expect

Situation	What you should see
Inputs missing	Amber labels; Calculate stays dim.
Calculate with valid inputs	RESULTS and the takeoff run; FLEX at its maximum by default.
TOW too high, CG out of envelope, wet runway...	A message with the reason; no result.
Send to FMS with a runway from the list	"Takeoff data sent to the FMS: RECEIVED CPNY T.O on the PERF page"; the MFD shows the data a moment later.
Send to FMS with a manual runway	Nothing is sent: the FMS needs the runway ident.
RECEIVED page, other runway than the flight plan	Runway in amber; INSERT not available.
INSERT	PERF T.O filled; THS FOR still to enter.

9. Limitations and why

Limitation	Why	What to do
Dry runways only.	The only public A380 takeoff data is the takeoff weight charts of the Airbus A380 Aircraft Characteristics (dry, ISA and ISA+15, runway 1700-5100 m, altitude 0-8000 ft).	On a wet runway, use the dry result with extra margin, as a training estimate only.
Speeds, FLEX, wind, slope and temperature effects are estimates (amber *).	The charts give weights only. The rest is a model fitted on them and checked on the FCOM examples (FLEX 38 at 560 t / 614 t).	-
The run distances (V1, VR, 35 ft points) are all-engine estimates.	No A380 distance data is public.	-
Send to FMS needs a runway from the list.	The FMS checks the uplink against its departure runway.	Choose the runway in the list.
Not certified data.	For simulation and training.	-